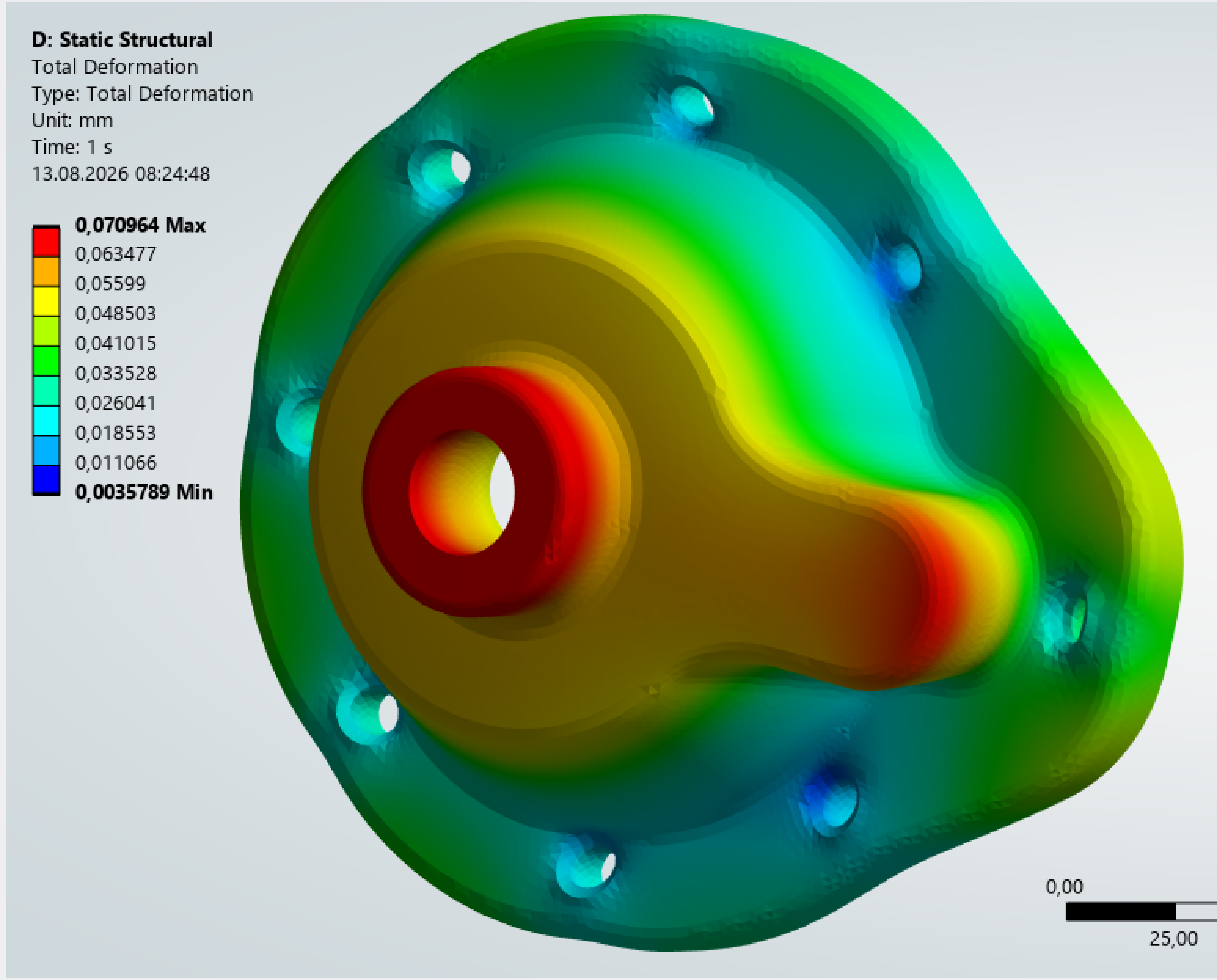
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Coupled Thermal–Structural Analysis of a Cast Iron Pump Housing Cover

Objective

Determine the steady-state temperature field, heat flux distribution and resulting thermal deformation of a gray cast iron pump housing cover operating between a hot pumped fluid and ambient air, and verify that the thermal solution conserves energy before any structural conclusion is drawn from it.

Model and Boundary Conditions

The cover is bolted to a pump body and separates the pumped fluid from the surrounding air. Three thermal boundary conditions were applied:

Boundary	Condition
Internal wetted surfaces (13 faces)	Fixed temperature, 82 °C (fluid)
Mating surface to pump body	Fixed temperature, 67 °C
External surfaces (32 faces)	Convection, stagnant air at 22 °C

The convection film coefficient was imported as a temperature-dependent correlation (stagnant air, simplified case) rather than entered as a constant, so the external heat transfer responds to the local surface temperature rather than being fixed in advance.

Material: gray cast iron, thermal conductivity 52 W/(m·K), density 7200 kg/m³. Mesh: tetrahedral, 4 mm global element size, resolution 7.

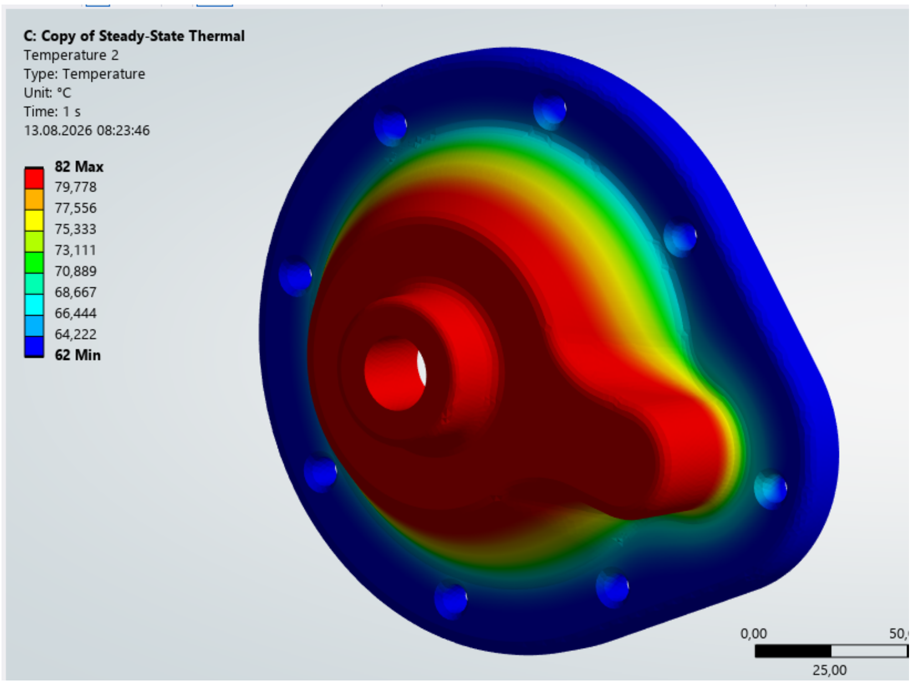
Verification: does the solution conserve energy?

Before reading any result, the thermal solution was checked against the first law. Reaction probes were placed on all three boundary conditions and the heat flows summed. A converged, correctly posed steady-state solution must close to zero:

Boundary condition	Heat flow
Internal surfaces at 82 °C (heat in)	+581.57 W
Mating surface at 67 °C (heat out)	–567.88 W
External convection (heat out)	–13.69 W
Net imbalance	–0.006 W

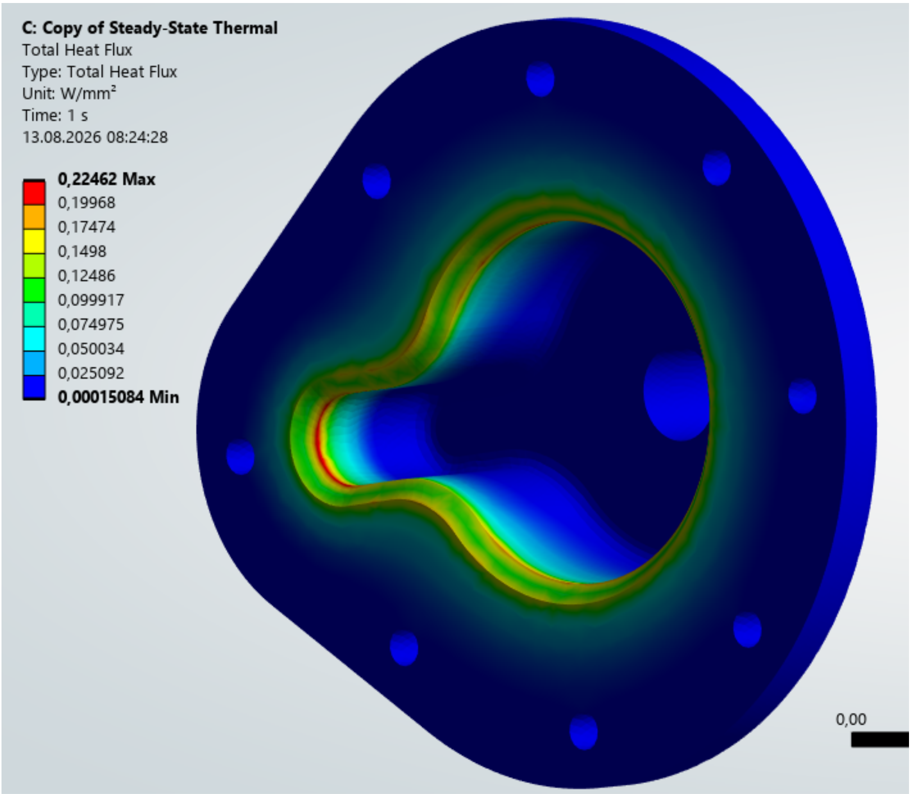
The residual is –0.006 W against 581.57 W entering the part, i.e. an energy imbalance of roughly 0.001%. The solution conserves energy, and the temperature field can be trusted as an input to the structural stage.

Thermal Results



Temperature distribution — 82 °C at the wetted core, falling to 62 °C at the flange rim

The temperature field spans 62–82 °C. The gradient is concentrated in a narrow band around the wetted cavity: the low thermal conductivity of gray cast iron confines the drop to the transition region rather than spreading it across the flange.

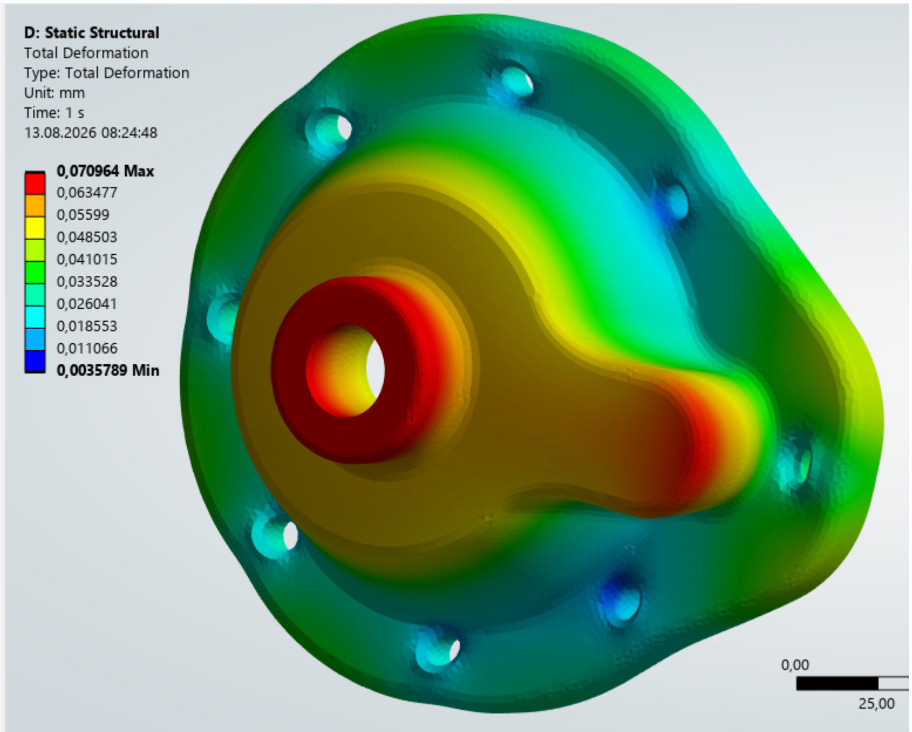


Total heat flux — peak 0.225 W/mm² at the cavity wall transition

Heat flux peaks at 0.225 W/mm² along the wall of the internal cavity, where the section is thinnest and the gradient steepest. The outer flange carries almost no flux, which is consistent with the convection probe accounting for only 13.69 W of the 581.57 W total — under these conditions the cover conducts heat into the pump body rather than dissipating it to ambient air.

Thermal Deformation

The temperature field was passed to a Static Structural analysis as an imported body load. Frictionless supports were applied to the bolt hole chamfers and the mating face, restraining normal displacement while allowing in-plane growth — a deliberately non-restrictive constraint set, since over-constraining a thermal expansion problem creates stresses that do not exist in the assembly.



Total deformation under thermal load — maximum 0.071 mm at the central boss

Maximum total deformation is 0.071 mm, located at the central boss — the hottest region and the point furthest from the restrained mating face. Deformation is smooth and monotonic, with no localised distortion, indicating free thermal growth rather than constrained expansion.

Note on peak stress: singularity, not a result

The equivalent stress plot from this model reports a maximum far above the material's ultimate strength. That value is not reported here as a result, because it is a numerical artefact rather than a physical stress: the peaks sit on the sharp edges of the bolt holes where the frictionless supports are applied. At a geometric singularity the computed stress does not converge — it keeps rising as the mesh is refined, without limit. Quoting such a peak as a design stress is one of the most common errors in FEA reporting.

For a design assessment the correct approach is to read stress away from the singular edges. A stress probe scoped to the flange rim, clear of the constrained hole edges, returns 115.3 MPa equivalent (von Mises) — roughly a factor of thirteen below the singular peak. That is the number a design decision can be based on; the peak is not.

Quantity	Value
Peak equivalent stress at constrained hole edges	singular — not reported
Equivalent stress in the flange body (probe)	115.3 MPa
Ultimate tensile strength, gray cast iron	240 MPa
Margin against tensile failure	2.1

A second point specific to this material: gray cast iron has no yield plateau. Its tensile yield strength is undefined, so a yield-based safety factor cannot be computed at all. Cast iron must be assessed against ultimate strength, and separately in tension and compression — 240 MPa versus 820 MPa in this dataset, a factor of 3.4. This is precisely why cast iron is used in compression-dominated housings and avoided where tensile loading governs.

Conclusions

– The thermal solution is verified: energy balance closes to 0.001% of the total heat input.

– Operating temperature range across the cover is 62–82 °C; peak heat flux 0.225 W/mm².

– Only 2.4% of the heat entering the cover leaves through external convection; the rest conducts into the pump body. External cooling is therefore not a meaningful design lever for this part.

– Maximum thermal deformation is 0.071 mm — small relative to typical flange sealing tolerances, so thermal growth is unlikely to compromise the gasket joint.

– Peak stresses at the constrained hole edges are singular and were excluded; stress in the flange body is 115.3 MPa against an ultimate tensile strength of 240 MPa, a margin of 2.1. A bolt-preload contact model would be required to qualify the hole regions themselves.